

CBMR Implementation Comparisons

Motivation, comparison setups and expanded findings

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Main finding. Numerical agreement remains strong and the improved aggregated-NB path is more stable at extreme parameters. The expanded benchmark does *not* show a consistent speed advantage over the original, already closed-form implementation. All 400 expanded calibration attempts were rejected by the final-score acceptance rule, despite optimiser success flags; consequently they provide no estimated coverage or false-positive rates.

This updates the earlier smoke-only performance and calibration summary. Sources are the saved JSON reports in `cbmr-proofs/comparison/results/`; experiments were not rerun to produce this document. The source hashes in both expanded reports match the current `cbmr_code/` and `cbmr_improved/` snapshots at generation.

1 Motivation and comparison setups

Exact SymPy identities and Lean proofs do not, by themselves, establish floating-point stability, successful optimisation, faster execution or reliable statistical inference. The comparison suite addresses these separate questions without modifying either implementation directory. GC denotes covariate effects constant across voxels; SV allows those effects to vary spatially. Covariate values may vary between studies in both designs.

Layer	Setup and purpose
Likelihoods and derivatives	Compare objectives, scores and Hessians at identical coefficients and fixed dispersion. Independent NumPy/SciPy closed formulas and unsimplified float64 PyTorch autodiff provide separate implementations; directional finite differences provide another check.
Models and designs	Poisson, aggregated NB and clustered NB are tested in both source packages. Independent-cell NB is reference/helper-only. GC and SV cases include shared covariates, unequal groups and varying exposures. Fixed quadratic penalties are external wrappers, not native package features.
Covariance	Compare block/Schur inversion, shared-global cross-group uncertainty, Kronecker structure and projected sandwiches against dense calculations. Include positive-definite, indefinite, nearly singular and singular cases.
Stability and fits	Exercise zero counts, extreme means, small dispersion and redundant intercepts. Identifiable, unpenalised fits use the same starting coefficients and L-BFGS-B configuration, with dispersion fixed.
Performance	Sequential fresh processes and one numerical thread. Time excludes imports and setup. Peak RSS is a whole-process high-water mark, not incremental operation memory.
Calibration	Simulate each model's own distribution and assess coefficient bias, 95% Wald coverage, null-moderator false positives and failed attempts separately from derivative agreement and timing.

Model boundary. Aggregated NB is a moment-matched pattern-total likelihood, not independent-cell NB. Its zero-dispersion limit is the *aggregate* Poisson objective. Global curvature need not equal full cell-level Poisson because study-allocation information is omitted.

2 Established numerical agreement and stability

2.1 Regression, covariance and complete fits

The saved `regression.json` records **512 required comparisons with zero failures**, plus 47 diagnostic observations. The standalone covariance and fit reports also have zero required failures (46 and 129 comparisons); these overlap the full regression report and must not be added to it.

- Across 312 derivative comparisons, the largest recorded absolute discrepancy was approximately 5.1×10^{-11} and the largest normalised Frobenius discrepancy was 3.1×10^{-12} .
- Covariance algebra agreed closely with dense calculations, with maximum absolute discrepancy approximately 1.4×10^{-14} . A nearly singular example had condition number approximately 6.9×10^{12} : small backward error does not guarantee accurate inverse entries. A redundant-intercept example exposed five identifiable dimensions for six coefficients.
- All 28 small implementation/model/design fits reported optimiser success; all 100 fit-agreement comparisons passed. The largest final score infinity norm was 4.23×10^{-6} , below the required 2×10^{-5} .

Most derivative comparisons require elementwise agreement with `rtol=1e-8`, `atol=1e-10`; finite-difference and fit checks use their own stated thresholds. These are finite numerical checks, not additional Lean proofs or statistical-calibration results.

2.2 Aggregated-NB stability

At spatial log means -1000 and $+1000$, the improved implementation returned finite objectives, scores and Hessians, whereas the original implementation and unsimplified reference produced nonfinite results. At dispersion 10^{-12} , the improved Hessian approached the correct aggregate-Poisson limit with maximum absolute error 9.45×10^{-11} , versus approximately 0.097 for the original and 1.34 for the reference. Autodiff is therefore not treated as an infallible floating-point oracle. Other difficult regimes are retained as diagnostics, rather than hidden behind the zero-failure headline.

3 Expanded performance experiment

3.1 What changed from the smoke benchmark

`benchmark-expanded.json` adds all three source likelihoods, two workloads and three problem sizes, using GC designs and three fresh-process repetitions per configuration. All 54 implementation/configuration records completed successfully: 162 measured workers plus 18 separate reference workers. The 18 size/model/workload cases are each run with the original, improved and unsimplified-autodiff implementations.

Size	Experiments	Voxels	Bases	Groups	Parameters
Small	32	32	4	3	13
Moderate	64	96	8	4	33
Large	128	256	8	4	33

Small-to-moderate scaling changes both data size and parameter count. Moderate-to-large keeps 33 parameters fixed. Even “large” is a bounded synthetic problem, not a whole-brain production benchmark.

3.2 Measured runtimes

All entries below are **median milliseconds across three repetitions**. The source implementations already contain closed-form Hessians; this tests the additional implementation changes, not merely closed forms versus autodiff. The autodiff benchmark uses a row-wise reverse-mode Hessian, not a reproduction of production forward-mode memory behaviour.

Hessian assembly.

Size	Model	Original	Improved	Autodiff	Ratio [†]
Small	Poisson	47.20	2.06	7.36	22.93×
Small	Aggregated NB	2.80	2.83	102.44	0.99×
Small	Clustered NB	1.38	97.50	410.00	0.01×
Moderate	Poisson	1.39	2.15	13.76	0.64×
Moderate	Aggregated NB	85.11	98.78	824.78	0.86×
Moderate	Clustered NB	15.67	16.90	442.64	0.93×
Large	Poisson	99.81	100.62	422.14	0.99×
Large	Aggregated NB	4.31	3.86	508.45	1.12×
Large	Clustered NB	1.64	2.39	196.93	0.69×

Contrast covariance. The same two-row contrast selects a group-intercept difference and a global moderator. The original route assembles information and a full covariance; the improved route assembles structured information and uses projected solves.

Size	Model	Original	Improved	Autodiff	Ratio [†]
Small	Poisson	2.12	3.18	7.38	0.67×
Small	Aggregated NB	3.46	4.03	115.39	0.86×
Small	Clustered NB	2.06	2.69	9.15	0.77×
Moderate	Poisson	1.99	96.95	347.60	0.02×
Moderate	Aggregated NB	102.32	98.13	719.02	1.04×
Moderate	Clustered NB	99.82	161.85	497.57	0.62×
Large	Poisson	2.07	3.10	107.28	0.67×
Large	Aggregated NB	84.32	5.01	491.25	16.81×
Large	Clustered NB	2.31	3.10	106.81	0.75×

[†]**Ratio = original median / improved median.** Above one means the improved route was faster; below one means it was slower. Ratios are descriptive measurements, not statistically established speedups.

3.3 Interpretation and limitations

The improved implementation was faster than the original in **4 of 18 paired cases**, and faster than the unsimplified-autodiff route in 18 of 18 cases. There is **no consistent additional speed advantage over the original** in this bounded experiment.

Numerical agreement held throughout: across original and improved benchmark outputs, the largest absolute discrepancy was 5.87×10^{-11} and the largest normalised Frobenius discrepancy was 8.94×10^{-15} . This supports implementation agreement in these cases, independently of which method happened to be faster.

The median whole-process RSS ranges were 259.5–264.4 MiB for the original and 258.7–262.9 MiB for the improved implementation. Interpreter, imports and setup dominate these totals; the results do not demonstrate meaningful operation-level or production-scale memory savings.

Only three repetitions were used, without warmup. For example, the improved moderate Poisson contrast workload took 2.46, 96.95, 197.17 ms in its three runs. This substantial variation limits timing conclusions and precludes a reliable scaling claim from these measurements alone.

4 Expanded calibration: no accepted inference replicates

`calibration-expanded.json` attempted **100 replicates per model** on the large GC design (128 experiments, 256 voxels, eight bases, four groups, 33 coefficients), for 400 attempts in total. Dispersion stayed fixed at its true value; all fits started from zero coefficients with the same L-BFGS-B protocol. Each model was simulated from its own specified sampling distribution.

Model	Optimiser success	Accepted fit / inference	Score norm minimum	Median	Maximum
Poisson	100	0 / 0	1.90×10^{-3}	6.12×10^{-3}	8.43×10^{-2}
Independent NB	100	0 / 0	2.00×10^{-3}	1.08×10^{-2}	5.82×10^{-2}
Aggregated NB	100	0 / 0	2.70×10^{-4}	1.07×10^{-3}	6.44×10^{-3}
Clustered NB	100	0 / 0	5.48×10^{-4}	2.60×10^{-3}	1.97×10^{-2}

Counts are out of 100 attempts per row. Score norm means $\|\nabla(-\ell)\|_\infty$. The acceptance rule requires a successful optimiser flag, finite results and score norm at most 1.00×10^{-4} before attempting inference.

4.1 Why the report says “completed with failures”

All 400 optimisations returned a success flag and terminated with the relative-objective-reduction message `REL_REDUCTION_OF_F_<=_FACTR*EPSMCH`. Nevertheless, **every final score norm exceeded the separate 1.00×10^{-4} requirement**. Thus there were 0 accepted fits and 0 accepted inference replicates. The recorded failure is an *acceptance-criterion failure*, not evidence that all optimisers crashed or that all information matrices were singular. Inference was not attempted after those rejections.

Coverage, false-positive rates and the report’s accepted-fit bias summaries are unavailable, not zero. Their denominators are zero and the corresponding JSON values are `null`. The run therefore cannot establish either good or poor statistical calibration. The earlier three-replicate-per-model smoke experiment had 12 accepted fits, but was far too small to support calibration claims; it does not override the expanded run’s acceptance failures.

4.2 What can be concluded and what remains to do

The expanded experiment strengthens the evidence for numerical agreement, but does not demonstrate a universal performance gain or validate inferential calibration. Before using the calibration experiment substantively, investigate why objective-reduction stopping precedes the pre-specified score requirement, validate an appropriate fitting/stopping protocol and rerun the simulations. Do not silently relax the rule, count rejected fits as successful, or interpret missing calibration estimates as zero error rates.

5 Provenance and reproducibility

Individual timings, optimiser diagnostics and rejected attempts remain in the saved JSON reports under `comparison/results/`. Report fingerprints (first 16 SHA-256 digits) are:

Report	Fingerprint
<code>benchmark-expanded.json</code>	<code>ef8cf087a589a267</code>
<code>calibration-expanded.json</code>	<code>77f57e0488d55b1a</code>

The editable source is `docs/explanation.tex`. `make explanation` regenerates its expanded-result tables from the saved JSON reports and rebuilds `explanation.pdf`; it does not rerun experiments. The generator checks source snapshots and the assumptions behind this interpretation, stopping for review if the experiment outcomes change.